Pythagoras - Mark scheme

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

Question 1 *Practice question written for this worksheet (not from a past paper)*

Is a triangle with sides 7 cm, 24 cm and 25 cm right-angled?

Show your working.

$7^2 + 24^2 = 49 + 576 = 625$

$25^2 = 625$, equal

Answer: Yes

Question 2 *Practice question written for this worksheet (not from a past paper)*

Is a triangle with sides 5 cm, 8 cm and 10 cm right-angled?

Show your working.

$5^2 + 8^2 = 89$

$10^2 = 100$, not equal

Answer: No

Question 3 *Practice question written for this worksheet (not from a past paper)*

A rectangle is 10 cm by 24 cm. Work out the length of its diagonal.

$10^2 + 24^2 = 676$

$\sqrt{676}$

Answer: 26 cm

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Question 4

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2021 · Question 14

Q	Working	Answer	Mark	Notes
	$30^2 + h^2 = 52^2$ oe or $900 + h^2 = 2704$ $(h^2 =) 52^2 - 30^2 (= 1804)$ or $(h^2 =) 2704 - 900 (= 1804)$		3	M1 for applying Pythagoras theorem correctly
	$(h =) \sqrt{52^2 - 30^2}$ $(= \sqrt{1804})$ $(= 42.47352..)$ or $(h =) \sqrt{2704 - 900}$ $(= \sqrt{1804})$ $(= 42.47352..)$			M1 for square rooting
	Correct answer scores full marks (unless from obvious incorrect working)	42.5		A1 awrt 42.5 or allow $2\sqrt{451}$
				Total 3 marks

Question 5

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2021 · Question 7

Question	Working	Answer	Mark	Notes
7	$\sin 42 = \frac{6.5}{x}$ **or** $\frac{x}{\sin 90} = \frac{6.5}{\sin 42}$ or $\cos 48 = \frac{6.5}{x}$ [where 48 = 180 – 90 – 42]		3	M1 **or** use of tan to find the horizontal side **and** then a correct first step in Pythagoras' theorem ie [base =] $\frac{6.5}{\tan 42} (= 7.21...)$ and [x =] 6.5 + "7.21…"
	[x =] $\frac{6.5}{\sin 42}$ **or** $\frac{6.5 \sin 90}{\sin 42}$ **or** [x =] $\frac{6.5}{\cos 48}$ [where 48 = 180 – 90 – 42]			M1 **or** complete method using Pythagoras $[x =] \sqrt{6.5^2 + "7.21..."^2}$ (If students give this statement with nothing before it they gain M2)
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	9.7		A1 accept 9.7 – 9.72
				Total 3 marks

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2020 · Question 7

7	$\tan x = \frac{3.4}{4.7}$ oe eg $\cos x = \frac{4.7}{\sqrt{3.4^2 + 4.7^2}}$ oe			M1 or $\sin x = \frac{3.4 \sin 90}{\sqrt{3.4^2 + 4.7^2}}$ oe
	$(x =) \tan^{-1}\left(\frac{3.4}{4.7}\right)$ oe eg $(x =) \cos^{-1}\left(\frac{4.7}{\sqrt{3.4^2 + 4.7^2}}\right)$			M1 or $(x =) \sin^{-1}\left(\frac{3.4 \sin 90}{\sqrt{3.4^2 + 4.7^2}}\right)$ oe
		35.9	3	A1 accept 35.7 - 36.1
				Total 3 marks

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2015 · Question 8

8	$(AB^2 =) 6.5^2 - 6.3^2$ (=2.56)		3	M1	**Alternative method :** M1 for finding a correct angle ($A = 75.7…$; $C = 14.2…$) **AND** a correct trig statement with a correct angle eg.
	$(AB =) \sqrt{6.5^2 - 6.3^2}$ **or** $\sqrt{"2.56"}$			M1 dep	$\sin 14.2 = \frac{AB}{6.5}$ M1 for making AB the subject eg. $AB = 6.5 \sin 14.2$
		1.6		A1	**NB: 1.6 as a rounded answer eg. from1.594… gains A0**
					Total 3 marks

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2025 · Question 20

Q	Working	Answer	Mark	Notes
	eg $(AC^2 =)18^2 - 7^2 (= 275)$ **or** $(AC =)\sqrt{18^2 - 7^2} (= \sqrt{275}$ or $5\sqrt{11}$ or $16.5(831...)$) **or** $(FB^2 =)9^2 + 7^2 (= 130)$ **or** $(FB =)\sqrt{9^2 + 7^2} (= \sqrt{130}$ or $11.4(017...)$) **or** $(GC^2 =)18^2 - 9^2 (= 243)$ **or** $(GC =)\sqrt{18^2 - 9^2} (= \sqrt{243}$ or $9\sqrt{3}$ or $15.5(884...)$) **or** $18^2 = (BC)^2 + 7^2 + 9^2$ oe		3	M1 for method to find AC^2 **or** AC **or** FB^2 **or** FB **or** GC^2 **or** GC **or** for a correct equation using BC^2 and 18 and 7 and 9 other longer ways to find AC, FB, GC may be used but must be a complete method eg $\angle FCA = \sin^{-1}\left(\frac{7}{18}\right) (= 22.88...)$ **and** $AC = \frac{7}{\tan"22.88..."}$
	eg $"275" - 9^2 (= 194)$ **or** $"16.5..."^2 - 9^2 (= 194)$ **or** $18^2 - "130" (= 194)$ **or** $18^2 - "11.4..."^2 (= 194)$ $"243" - 7^2 (= 194)$ **or** $"15.5..."^2 - 7^2 (= 194)$ **or** $18^2 - 7^2 - 9^2 (= 194)$ **or** $\angle FCB = \sin^{-1}\left(\frac{"11.4"}{18}\right)$ $(= 39.3(036...))$ **and** $\cos"39.3" = \frac{(BC)}{18}$ or $\tan"39.3" = \frac{"11.4"}{(BC)}$ oe			M1 for complete method to find BC^2 other longer ways to find BC may be used but must be a complete method, leading to a trig equation in BC
	Correct answer scores full marks (unless from obvious incorrect working)	13.9		A1 accept 13.8 to 14
				Total 3 marks

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Question 9

Practice question written for this worksheet (not from a past paper)

A right-angled triangle has sides x, $x+1$ and $x+2$ (the hypotenuse is $x+2$).

Show that $x^2 - 2x - 3 = 0$ and work out x.

$x^2 + (x+1)^2 = (x+2)^2$

$2x^2 + 2x + 1 = x^2 + 4x + 4$, so $x^2 - 2x - 3 = 0$

$(x-3)(x+1) = 0$; $x > 0$

Answer: x = 3

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · November 2020 · Question 7

7	$4.3^2 + 6.4^2$ or 59.45		4	M1	for squaring and adding
	$\sqrt{4.3^2 + 6.4^2}$ or $\sqrt{59.45}$ or 7.71(038…) or 7.7			M1	dep 1st M1 for square rooting
	e.g ('7.71' + 4.3 + 6.4) × 22 or '18.4' × 22 (=404.8) or ('8' + 4.3 + 6.4) × 22 or '18.7' × 22 or '19' × 22 or '20' × 22			M1	dep M2 for a non-rounded perimeter × 22 or 19 × 22 accept 20 × 22 oe
		\$418		A1	cao
					Total 4 marks

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2023 · Question 20

Q	Working	Answer	Mark	Notes	
	$7.2^2 + 5.4^2$ (= 81)		4	M1 for correct first step using Pythagoras	M1 for reaching one step from the length of AB if using trig eg $(EAB =)\tan^{-1}\left(\frac{5.4}{7.2}\right)(=36.8...)$ and $\sin("36.8...") = \frac{5.4}{AB}$
	$\sqrt{7.2^2 + 5.4^2}$ (= 9)			M1 for complete Pythagoras method to find length of AB/DC check the diagram for sight of 9, DC marked as 9 implies M2	M1 for complete method to find the length of AB/DC eg $\frac{5.4}{\sin("36.8...")}(=9)$
	7.2 + 5.4 + 6 + "9" + 6 oe			M1 for a complete method to find the perimeter	
	Correct answer scores full marks (unless from obvious incorrect working)	33.6		A1 oe	
				Total 4 marks	

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2025 · Question 11

11	$240=\frac{1}{2}\times 30\times(AC)$ or $240=\frac{1}{2}\times 30\times(AC)\times\sin 90$ or $(AC=)\frac{240\times 2}{30}(=16)$ or $(AC=)\frac{240\times 2}{30\times\sin 90}(=16)$		4	M1
	$(BC^2=)30^2+\text{“16”}^2\ (=1156)$ or $(BC^2=)30^2+\text{"16"}^2-2\times 30\times 16\times\cos 90\ (=1156)$ or $(ABC=)\tan^{-1}\left(\frac{\text{"16"}}{30}\right)(=28.0(724...))$ or $(ACB=)\tan^{-1}\left(\frac{30}{\text{"16"}}\right)(=61.9(275...))$			M1 dep on M1 ft their AC value provided the first M1 has been awarded
	$(BC=)\sqrt{30^2+\text{"16"}^2}\ (=\sqrt{1156}=34)$ or $(BC=)\sqrt{30^2+\text{"16"}^2-2\times 30\times 16\times\cos 90}\ (\sqrt{1156}=34)$ or $(BC=)\frac{\text{"16"}}{\sin(\text{"28.0"})}(=34)$ or $(BC=)\frac{30}{\cos(\text{"28.0"})}(=34)$ or $(BC=)\frac{\text{"16"}}{\cos(\text{"61.9"})}(=34)$			M1
	Correct answer scores full marks (unless from obvious incorrect working)	80		A1 Allow 79.9 – 80.1 SCB2 for 69(.048…) SCB1 for 31(.048…) – 31.1 or $\sqrt{964}$ or $2\sqrt{241}$
				Total 4 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 11

11	$(CM)^2+(12\div 2)^2=9^2$ oe or $9^2-(12\div 2)^2\ (=81-36=45)$		4	M1 $AM = MB$ $CAM = CBM$	M2 for $(\cos^{-1}(CAM)=)\frac{12\div 2}{9}=48.1(896...)$
	$\sqrt{9^2-(12\div 2)^2}$ oe $(=\sqrt{81-36}=\sqrt{45}=3\sqrt{5}=6.7(08...))$			M1	**and** $(CM=)(12\div 2)\times\tan\text{"48.1..."}(=6.7…)$ or $(CM=)9\times\sin\text{"48.1..."}(=6.7…)$
	(“7”+ 9 + 9 + 12) × 21.5(0) or 37 × 21.5(0)			M1	
	Working required	795.5(0)		A1 dep on M2 SC B3 for awrt 789 for using 6.7…	
					Total 4 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2013 · Question 11

11.	2.1 – 1.7 (= 0.4) $6^2+\text{“0.4”}^2\ (=36.16)$ √ “36.16”	6.01	4	M1 M1 dep M1 dep A1 awrt 6.01 N.B. Accept working in cms throughout for method marks
				Total 4 marks

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · January 2023 · Question 24

Q	Working	Answer	Mark	Notes
	$\frac{1}{2}\times 7\times h = 42$ oe or $(h=)\frac{42\times 2}{7}$ $(=12)$ oe or $3.5^2+h^2=y^2$ or $h=\sqrt{y^2-3.5^2}$ oe		4	M1 A correct equation involving the height or a correct expression for height – could be in terms of y
	$y^2=\left(\frac{7}{2}\right)^2+("12")^2$ oe or $\frac{1}{2}\times 7\times "\sqrt{y^2-3.5^2}" = 42$ oe			M1 (indep) use of *their* height (any found value that they have called 'height')
	$y=\sqrt{\left(\frac{7}{2}\right)^2+("12")^2}$ oe			M1 **all values must come from a correct method**
	Correct answer scores full marks (unless from obvious incorrect working)	12.5		A1 oe eg $\frac{25}{2}$
				Total 4 marks

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2022 · Question 25

Q	Working	Answer	Mark	Notes
(a)	$(18\ 3-\ +\ --)^2\ (7\ 1)^2$oe or $15^2+\ =8^2(\ 289)$ oe		3	M1
	$\sqrt{(18-3)^2+(7--1)^2}$ $(=\sqrt{"289"})$			M1
		17		A1
(b)	$13+6>$ "17"	correct reason	1	A1ft dep M1 **Acceptable examples** "They overlap by 2cm" "The distance between the centres is less than the sum of the radii" "17 is less than the distance than the total of the radii" "19 is bigger than the distance between the centres" **Not acceptable examples** "19 is greater than the distance between the circles" oe "The circumference of each circle overlaps"
				Total 4 marks

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Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2022 · Question 18

18	eg $(BV^2 =)3^2+6^2(=45)$ **or** $(CT^2 =)3^2+6^2(=45)$ **or** $(DH^2 =)\ 6^2+6^2\ (=72)$ **or** $(MV^2 =)3^2+3^2(=18)$		4	M1	a correct expression for eg BV^2 **or** CT^2 **or** DH^2 **or** MV^2 where M is the midpoint of DC **or** a correct expression for [length]^2 for any length in the cube using Pythagoras	M3 for $(VT =)\sqrt{6^2+3^2+3^2}$ $(=3\sqrt{6}$ or 7.34...) (M2 for $(VT^2 =)$ $6^2+3^2+3^2\ (=54)$)
	eg $(BV =)\sqrt{3^2+6^2}\ \left(=\sqrt{45}\text{ or }3\sqrt{5}\text{ or }6.70...\right)$ **or** $(CT =)\sqrt{3^2+6^2}\ \left(=\sqrt{45}\text{ or }3\sqrt{5}\text{ or }6.70...\right)$ **or** $(DH =)\sqrt{6^2+6^2}\ \left(=\sqrt{72}\text{ or }6\sqrt{2}\text{ or }8.48...\right)$ **or** $(MV =)\sqrt{3^2+3^2}\ \left(=\sqrt{18}\text{ or }3\sqrt{2}\text{ or }4.24...\right)$			M1	for a complete method for eg BV **or** CT **or** DH **or** MV **or** any length in the cube using Pythagoras	
	$(VT =)\ \sqrt{"45"+3^2}$ **or** $\sqrt{\left(\frac{"\sqrt{72}"}{2}\right)^2+6^2}$ **or** $\sqrt{"18"+6^2}$ **or** $3\sqrt{6}$ **or** 7.34...			M1	for a correct expression for VT (condone missing brackets around $3\sqrt{5}$ or $3\sqrt{2}$ or $\frac{\sqrt{72}}{2}$)	
		$\sqrt{54}$		A1	if $\sqrt{54}$ seen and answer then given as $3\sqrt{6}$ isw and award full marks	
						Total 4 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2015 · Question 23

23	Let O be the centre of the square. $(AC^2) = 10^2 + 10^2\ (=200)$ **or** $(AC =)\ \sqrt{200}$ oe **or** $(AC =)\ 14.1(4\ldots)$		4	M1 or $2AO^2 = 10^2$
	$(AO =)\ \frac{1}{2}\sqrt{200}$ oe **or** $(AO) = 7.07(1\ldots)$ **or** $(AO) = 7.05$			M1
	$\left(VO^2 =\right) 12^2 - \left(\frac{1}{2}\sqrt{200}\right)^2$ oe $(=94)$ **OR** Angle VAC is $\cos^{-1}\left(\frac{7.07}{12}\right) = 53.896^\circ$ **AND** 12 sin 53.896 (= 9.695...)			M1 (dep on both previous method marks) for a fully correct method (condone missing brackets)
		9.70		A1 awrt 9.70 accept 9.7
	Alternative method Let M be the midpoint of a side of the square $VM^2 = 12^2 - 5^2\ (=119)$ **or** $VM = \sqrt{119}$ (=10.9(…)			M2 **but it must be explicitly clear that it is *VM* being calculated**
	$VO^2 = 119 - 5^2\ (=94)$ **or** $VO^2 = 10.9^2 - 5^2$ oe			M1
		9.70		A1 awrt 9.70 accept 9.7
				Total 4 marks

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Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2018 · Question 9

9	E.g. $\tan 72 = \frac{12.8}{a}$ **or** $\tan(90-72) = \frac{o}{12.8}$ **or** $\sin 72 = \frac{12.8}{h}$ **or** $\cos(90-72) = \frac{12.8}{h}$	110	5	M1	substitutes correctly into a trig ratio (including the Sine rule)
	E.g.(shortest side) = $\frac{12.8}{\tan 72}$ **or** 12.8tan(90 – 72) **or** 4.15(89…) **or** 4.16 **or** (hypotenuse =) $\frac{12.8}{\sin 72}$ **or** $\frac{12.8}{\cos(90-72)}$ **or** 13.4(58…) **or** 13.5			M1	for a complete method to find one side of the triangle
	One of (shortest side =) $\frac{12.8}{\tan 72}$ **or** 12.8tan(90 – 72) **or** 4.15(89…) **or** 4.16 **or** $\sqrt{"13.4..."^2 - 12.8^2}$ **AND** **One of** (hypotenuse =) $\frac{12.8}{\sin 72}$ **or** $\frac{12.8}{\cos(90-72)}$ **or** 13.4(58…) **or** 13.5 **or** $\sqrt{12.8^2 + "4.15..."^2}$			M1	for a complete method to find both missing sides of triangle NB Could use Pythagoras's theorem with side found – must be a complete correct method
	5 × ("13.4(58…)" – "4.15(89…)") + 5 × 12.8 **or** 5 × ("13.4…" + "4.15…" + 12.8) – 10 × "4.15…"			M1	for method to use found lengths to find perimeter
				A1	for answer in range 110 - 111

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 10

Q	Working	Answer	Mark	Notes
10	$(AC^2 =)12^2 + 16^2 (= 144 + 256 = 400)$ **or** $(BAC =)\tan^{-1}\left(\frac{12}{16}\right)(= 36.8(698...))$ **or** 36.9 **or** $(BCA =)\tan^{-1}\left(\frac{16}{12}\right)(= 53.1(301...))$		5	M1 for a correct method using triangle *ABC*
	$(AC =)\sqrt{12^2 + 16^2}\left(= \sqrt{144 + 256} = \sqrt{400} = 20\right)$ **or** $(AC =)\frac{16}{\cos"36.8"}(= 20)$ **or** $(AC =)\frac{12}{\sin"36.8"}(= 20)$ **or** $(AC =)\frac{16}{\sin"53.1"}(= 20)$ **or** $(AC =)\frac{12}{\cos"53.1"}(= 20)$			M1 for a correct method to find *AC*
	$(BD^2 =)(1.5\times"20")^2 - 16^2 (= 644)$ **or** $(BD^2 =)30^2 - 16^2 (= 900 - 256 = 644)$ **or** $(BAD =)\cos^{-1}\left(\frac{16}{"30"}\right)(= 57.7(690...))$ **or** 57.8 **or** $(BDA =)\sin^{-1}\left(\frac{16}{"30"}\right)(= 32.2(309...))$ **or** $(BCA =)\sin^{-1}\left(\frac{16}{"20"}\right)(= 53.1(301...))$ **and** $\frac{(CD)}{\sin(180-126.9-32.2)} = \frac{"30"}{\sin(180-53.1)}$ oe			M1 for a correct method using triangle to find BD^2 or angle *BAD* or angle *BDA* or for a correct equation for side *CD*
	$(BD =)\sqrt{(1.5\times"20")^2 - 16^2}$ (= 25.3(771)…) **or** $(BD =)\sqrt{30^2 - 16^2}\left(= \sqrt{900 - 256} = \sqrt{644} = 2\sqrt{161} = 25.3(771...)\right)$ **or** $(BD =)16\times\tan"57.7"(= 25.3(771...))$ **or** $(BD =)"30"\times\sin"57.7"(= 25.3(771...))$ **or** $(BD =)\sqrt{16^2 + "30"^2 - 2\times16\times"30"\times\cos"57.7"}(= 25.3(771...))$ **or** $(BD =)30\times\cos"32.2"(= 25.3(771...))$ **or** $(BD =)\frac{16}{\tan"32.2"}(= 25.3(771...))$ **or** $(BD =)\frac{16}{\sin"32.2"}\times\sin"57.7"(= 25.3(771...))$ **or** $(CD =)\frac{"30"}{\sin"126.9"}\times\sin"20.9"$ oe			M1 for a correct method to find *BD* or *CD*
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	13.4		A1 awrt 13.4
				Total 5 marks

Question 21 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 10*

Question	Working	Answer	Mark	Notes
10	$(54-24)\div 2\ (=15)$ [may be marked on diagram]		5	M1
	$"15"^2-(24\div 2)^2\ (=81)$			M1 ft their "15" (if > 12)
	[height =] $\sqrt{"15"^2-(24\div 2)^2}\ (=9)$			M1 ft their "15" (if > 12)
	$(24\times "9")\div 2$ oe			M1 figures must be from correct working
	Correct answer scores full marks (unless from obvious incorrect working)	108		A1 allow 107.9 – 108.1
	ALTERNATIVES BELOW			**Total 5 marks**
10	$(54-24)\div 2\ (=15)$ [may be marked on diagram]		5	M1
	or $x=\cos^{-1}\left(\frac{"12"}{"15"}\right)(=36.86...)$ **or** $y=\sin^{-1}\left(\frac{24\div 2}{"15"}\right)(=53.13...)$ **or** $A=\cos^{-1}\left(\frac{15^2+15^2-24^2}{2\times 15\times 15}\right)(=106.2...)$ **or** $B=\cos^{-1}\left(\frac{15^2+24^2-15^2}{2\times 15\times 24}\right)(=36.8....)$			M1 ft their "15" (if > 12) [using Hero's formula S = 0.5 × 54 (= 27) and] 27 × (27 – 24) × (27 – "15") × (27 – "15")
	or "12"tan"36.86…" (= 9) (allow 8.9… for these) "12" ÷ tan"53.13…" (= 9) **or** "15" × sin "36.86…" (= 9) **or** "15" × cos "53.13…" (= 9)			M1 ft their "15" (if > 12) M2 for 0.5 × 24 × "15" × sin"36.86…" or 0.5 × "15" × "15" × sin(2 × "53.13…") or 0.5×"15"×"15"×sin("106.2...") **or** $\sqrt{"27"("27"-24)("27"-"15")("27"-"15")}$
	$(24\times "9")\div 2$ oe			M1
	Correct answer scores full marks (unless from obvious incorrect working)	108		A1 allow 107.9 – 108.1
				Total 5 marks

Question 22 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2024 · Question 23*

Q	Working	Answer	Mark	Notes
	$\frac{14+AB}{2}\times 15=360$ oe **or** $360-14\times 15(=150)$ oe		6	M1 for setting up an equation using the area of the trapezium **or** method to find the area of the triangle
	$AB=34$ **or** $MB=20$ (where M is point on AB such that MC is perpendicular to AB)			A1 could be seen on diagram
	$(CB^2=)15^2+20^2\ (=625)$ **or** $(CB^2=)15^2+MB^2$			M1 allow use of their MB
	$(CB=)\sqrt{15^2+20^2}\ (=25)$ **or** $(CB=)\sqrt{15^2+MB^2}$			M1 allow use of their MB
	14 + 15 + "34" + "25" oe **or** 14 + 15 + 14 + MB + CB oe			M1ft (dep on previous two M marks) for a method to find the perimeter of the trapezium, allow use of their MB and CB
	Correct answer scores full marks (unless from obvious incorrect working)	88		A1 cao
				Total 6 marks

CHALLENGE QUESTIONS

Question 23 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2022 · Question 27*

Q	Working	Answer	Mark	Notes
	$(AB^2 =)\ 7.5^2 - 6^2\ (= 20.25)$ **or** eg $(BAC =)\sin^{-1}\left(\frac{6}{7.5}\right)(= 53.1...)$ **or** $\cos(BCA) = \frac{6}{7.5}(= 0.8)$		6	M1 for a correct first step to find AB **or** a complete method to find angle BAC **or** a correct first step to find angle BCA
	$(AB =)\ \sqrt{7.5^2 - 6^2}\ (= 4.5)$ **or** $(AB =)\frac{6}{\tan"53.1"}(= 4.5...)$ **or** $(AB =)7.5\cos"53.1"(= 4.5...)$ **or** $(BCA =)\cos^{-1}\left(\frac{6}{7.5}\right)(= 36.8...)$			M1 for a complete method to find AB **or** angle BCA
	$(\text{Area } ABC =)\ \frac{1}{2}\times 6\times"4.5"\ (= 13.5)$ **or** $(\text{Area } ABC =)\frac{1}{2}\times 6\times 7.5\times\sin("36.8")$ $(= 13.47...$ or $13.5)$			M1 ft [their labelled AB] or [their labelled BCA] eg for $\frac{1}{2}\times 6\times$ [their labelled AB] **or** $\frac{1}{2}\times 6\times 7.5\times\sin$ [their labelled BCA]
	$(\text{Area } DAC =)\ 31.5 - "13.5"\ (= 18)$ **or** $"13.5" + 0.5 \times 7.5 \times AD = 31.5$ oe			M1 ft (dep on previous M1) allow 31.5 – [their area]
	$(AD =)\ ("18" \div 7.5) \div 0.5$ oe			M1 for a complete method to find AD, dependent on correct working
		4.8		A1 accept 4.78 – 4.81
				Total 6 marks

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Question 24

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2023 · Question 20

	eg $(AB^2 =)6^2 + 6^2 (= 72)$ or $\sin 45 = \frac{6}{(AB)}$ or $\cos 45 = \frac{6}{(AB)}$ or or $(AB^2 =)6^2 + 6^2 - 2\times6\times6\times\cos 90$		5	M1 for a correct start to the method to find AB
	eg $(AB =)\sqrt{6^2 + 6^2}\ (= \sqrt{72}$ or $6\sqrt{2}$ or $8.48...)$ or $(AB =)\frac{6}{\sin 45}\ (= \sqrt{72} = 6\sqrt{2} = 8.48...)$ or $(AB =)\frac{6}{\cos 45}\ (= \sqrt{72} = 6\sqrt{2} = 8.48...)$ or $(AB =)\sqrt{6^2 + 6^2 - 2\times6\times6\times\cos 90}$			M1 for a complete method to find the length of AB
	eg $\pi\times6\ (= 6\pi$ or $18.8...)$ or $\pi\times6\div2\ (= 3\pi$ or $9.42...)$ or $\pi\times$"8.48..."$(= 26.6...)$ or $\pi\times$"8.48..."$\div 2\ (= 13.3...)$			M1 (indep) for a method to find the circumference of one whole circle or the arc length of one semicircle seen (may be embedded)
	eg $2\times$"3π"+"13.3..." or "9.42" + "9.42" + "13.3" or "18.8" + "13.3"			M1 for a complete correct method to find the perimeter of the shape
	Correct answer scores full marks (unless from obvious incorrect working)	32.2		A1 accept answers in the range 32.1 – 32.3
				Total 5 marks

Question 25

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2024 · Question 10

Q	Working	Answer	Mark	Notes
	eg $51^2 = (DE)^2 + 24^2$ oe or $2601 = (DE)^2 + 576$ oe or $(DE^2 =)51^2 - 24^2\ (= 2025)$ oe or $(DE^2 =)2601 - 576\ (= 2025)$ oe or $\cos(DFE) = \frac{24}{51}$ or $\sin(DEF) = \frac{24}{51}$		5	M1 for applying Pythagoras theorem correctly
	$(DE =)\sqrt{51^2 - 24^2}\ (= \sqrt{2025} = 45)$ or $(DE =)\sqrt{2601 - 576}\ (= \sqrt{2025} = 45)$			M1 for square rooting

	$(DE=)\sqrt{2601-576}\ (=\sqrt{2025}=45)$ **or** $(DFE=)\cos^{-1}\left(\frac{24}{51}\right)(=61.9...)$ **or** $(DEF=)\sin^{-1}\left(\frac{24}{51}\right)(=28.0...)$			
	$\frac{\text{their } DE}{7.5}(=6)$ oe **or** $\frac{7.5}{\text{their } DE}\left(=\frac{1}{6}\right)$ **or** $\frac{x}{24}=\frac{7.5}{\text{their } DE}$ oe **or** tan"their 61.9..." $=\frac{7.5}{(x)}$ **or** tan"their 28.0..." $=\frac{(x)}{7.5}$ **or** $\frac{(x)}{\sin(\text{their } 28.0)}=\frac{7.5}{\sin(\text{their } 61.9)}$ NB Their *ED* or their 61.9 or their 28.0 must be clearly identified Their 61.9.. or their 28.0.. cannot be used as lengths of the triangle Their 45 cannot be used as an angle of the triangle			M1 for a correct method to find the scale factor Allow correct use of sine rule/cosine rule/Pythagoras theorem Allow 0.17 or better for $\frac{1}{6}$ **Special case** Allow $(DE=)\sqrt{51^2+24^2}$ $(=\sqrt{3177}=3\sqrt{353}=56.3...)$ for "45" for this mark
	"6" oe **or** $24\times"\frac{1}{6}"$ **or** 24 × "1.67" **or** $(x=)\frac{7.5}{\text{their } ED}\times 24$ **or** $(x=)\frac{7.5}{\tan(\text{their } 61.9...)}$ oe **or** $(x=)7.5\times\tan(\text{their } 28.0...)$ oe **or** $(x=)\frac{7.5}{\sin(\text{their } 61.9)}\times\sin(\text{their } 28.0)$ oe **or** $51\times"\frac{1}{6}"(=8.5)$ **and** $(x=)\sqrt{8.5^2-7.5^2}\left(=\sqrt{16}\right)$			M1 dep on previous M1 for a correct method to find x or for finding BC **and** using Pythagoras theorem to find x Allow $24\times"\frac{7.5}{56(.3...)}"$ **or** $24\div"\frac{56(.3...)}{7.5}"$ for scale factor for this mark
	Working required	4		A1 dep on M2 The value of 4 must come from correct figures
				Total 5 marks

Question 26

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 11

11	$(AD^2 =)\ 28^2 + 45^2\ \ (=784 + 2025 = 2809)$ **or** $(EDA =)\ \tan^{-1}\left(\frac{28}{45}\right)\ \ (=31.8(9...))$ **or** $(EAD =)\ \tan^{-1}\left(\frac{45}{28}\right)\ \ (=58.1(09...))$		5	M1	for a correct method to find *AD* **or** angle *EDA* **or** angle *EAD*
	eg $(AD =)\ \sqrt{28^2 + 45^2}\ \ (=\sqrt{784 + 2025} = \sqrt{2809} = 53)$ oe **or** $(AD =)\ \frac{28}{\sin 31.8(9...)}\ \ (=53...)$ **or** $(AD =)\ \frac{45}{\cos 31.8(9...)}\ \ (=53...)$ oe **or** $(AD =)\ \frac{45}{\sin 58.1...}\ \ (=53...)$ **or** $(AD =)\ \frac{28}{\cos 58.1...}\ \ (=53...)$ oe			M1	for a correct method to find *AD*
	eg $(BC =)\ \sqrt{35^2 - ("53" - 21)^2}\ \ (=\sqrt{1225 - 1024} = \sqrt{201} = 14.1(7...))$ **or** $(BDC =)\ \cos^{-1}\left(\frac{"53" - 21}{35}\right)\ (= 23.8(9...))$ **or** $(BCD =)\ \sin^{-1}\left(\frac{"53" - 21}{35}\right)\ (= 66.1...)$			M1	for correct method to find *BC* **or** angle *BDC* **or** angle *BCD*
	eg $\frac{1}{2} \times "14.1(7...)" \times ("53" - 21)\ \ (= 16\sqrt{201})$ **or** $\frac{1}{2} \times 35 \times ("53" - 21) \times \sin("23.8(9...)")$ **or** $\frac{1}{2} \times 35 \times "14.1(7...)" \times \sin("66.1...")$			M1	for a correct method to find the area of triangle *CDB*
	Correct answer scores full marks (unless from obvious incorrect working)	227		A1	awrt 227 accept $16\sqrt{201}$
					Total 5 marks

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Question 27

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 28

Question	Working	Answer	Mark	Notes
	$\frac{1}{2}\pi r^2 = 227$ oe **or** $\pi r^2 = 227\times 2$ oe **or** $r^2 = 144.5...$ **or** $r = \sqrt{144.5...}\ (=12(.02...))$		5	M1 for a correct equation or a correct calculation to find the radius of the semicircle
	$\frac{2\pi\times"12"}{2}$ oe **or** $\pi\times"12"\ (=37.6(99...))$ **or** $\pi\times"12.02..."\ (=37.7(661...))$ **or** $\pi\times$[their radius]			M1 for a method to find the arc length of the semicircle ft their value for the radius as long as clearly identified (may be on the diagram or in the working)
	$\sqrt{15^2-"12"^2}\ (=9)$ **or** $\sqrt{15^2-"12.02..."^2}\ (=8.9(7...))$ $\sqrt{15^2-[\text{their radius}]^2}$			M1 for a complete method to find the height of the right angled triangle ft their value for the radius as long as clearly identified (may be on the diagram or in the working)
	"12"π + 2×15 + 2×"9" (= 85.6(99...)) **or** "12.02..."π + 2×15 + 2×"8.97..." (= 85.7(09...)) **or** "12"π + 48 **or** "12.02..."π + 48 **or** "37.6(99...)" + 15 + 15 + "8.97..." + "8.97..." (= 85.7(09...)) **or** [arc length] + 2×15 + 2×[*AB* or *DE*]			M1 dep on M2 for a method to find the perimeter including at least the arc length **and** one length of 15 **and** one of *AB* or *DE* ft their values for the arc length and *AB* or *DE* as long as clearly identified (may be on the diagram or in working) allow the error of addition of the radius (*OB* or OD) or the diameter *BD*
	Working required	85.7		A1 dep on M2 accept an answer in the range 85.6 – 85.8
				Total 5 marks

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